MEDICAL CORE · RADIOTHERAPY RESEARCH

Surface-guided Respiratory Tracking (SGRT)

Build the optical part of surface-guided radiotherapy - patient-surface setup registration and respiratory gating - on a domestic 3D sensor stack.

EVIDENCE STATE	PERIOD
Ongoing · Research	2026.06 - present

PORTFOLIO TIER	PRIMARY CAPABILITY
Medical Core	3D Geometry & Registration

An early-stage research assignment contracted to DIGITRACK within the K-LINAC programme (led by KERI, imaging sub-project led by ETRI): far-field surface reconstruction and near-field real-time breathing tracking with commercial 3D sensors. Commercial systems do not publish the reasoning behind their sensor and algorithm choices, so a domestic stack has to build that evidence itself.

Problem and system boundary

01 / TWO OPTICAL TRACKS

Two optical tracks

Far-field sensors merge point clouds into a surface and register it to the planning CT; near-field sensors turn chest and abdomen ROI depth into a respiratory waveform and gating signal.

PROBLEM

Patient position and breathing must be known during treatment without extra imaging or skin marks.

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PUBLIC BOUNDARY

First-year sensor validation only; no clinical performance or programme-target achievement is claimed, and programme targets, budgets, and metrics of other institutions are not published here.

The clinical partner is the radiation oncology department of Seoul St. Mary's Hospital; interfaces are agreed with 4DCT reconstruction, image guidance, and integrated-control partners.

EVIDENCE STATE

Ongoing · Research / The in-house validation tool DtDepthScan measuring a silicone body phantom: mean depth and temporal noise σ are computed live from the ROI on the point cloud. Device serial and address are masked.

Owned decisions and implementation

MY ROLE

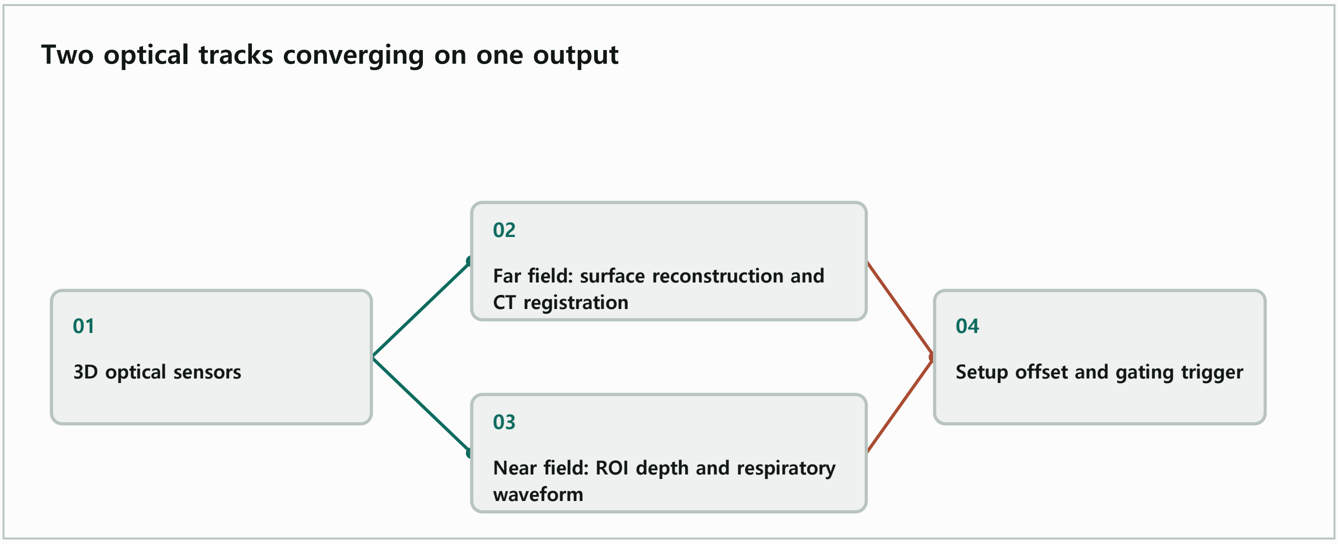
Lead the sensor validation campaign, which first required building the in-house tool DtDepthScan (Qt, VTK, OpenCV): raw continuous recording with acquisition timestamps, ROI depth time series, automatic σ, delivered fps and fill rate, and PLY/CSV export for cross-checking elsewhere. Then design and implement the breathing-tracking algorithm from ROI depth to respiratory waveform and gating signal, define the sensor interface and transport protocol, and run day-to-day project work.

02 / THE TOOL CAME FIRST

The tool came first

Abstracting the camera layer lets any sensor run the same procedure. Timestamped raw recording gives delivered fps, ROI depth time series yield σ and fill rate automatically, and PLY and CSV export allow cross-checking elsewhere.

SYSTEM RELATIONSHIP



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

ToF camera / Structured light / Qt / VTK / OpenCV / Python / 4DCT

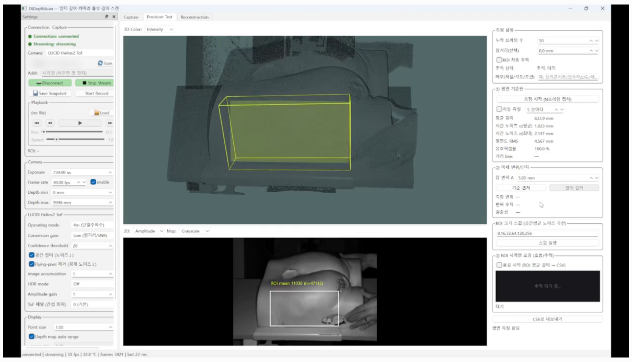
Verification and evidence ledger

03 / ONE PROCEDURE, FIVE SENSORS

One procedure, five sensors

Silicone body phantom, five distances 0.5-3.0 m · Five sensors and condition sweeps, 55 runs · 500 frames per run · Repeatability ±0.05 mm, σ settles by 50 frames

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-lead-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-gallery-04

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-gallery-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-gallery-05

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-gallery-02

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-gallery-06

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

respiratory-surface-guidance-gallery-03

Approved derivative; caption in portfolio data.

VERIFICATION AND EVIDENCE LEDGER

A silicone body phantom measured at five distances from 0.5 to 3.0 m with five commercial 3D sensors, 55 runs of 500 frames each, gave precision σ, delivered fps, and fill rate; repeats agreed to within ±0.05 mm. That table and the DtDepthScan captures are my own measurements.

Role, team result, and limits

04 / THE CONDITIONS OUTLIVED THE NUMBERS

The conditions outlived the numbers

The operating conditions will outlast the σ table. At 0.5 m the default exposure saturates the IR return and no depth is produced, and from a cold start σ is up to 1.7 times worse than at thermal steady state. Both became design inputs for the real-time module.

MY ROLE

Own sensor validation, tooling, tracking algorithm, and protocol.

Lead the sensor validation campaign, which first required building the in-house tool DtDepthScan (Qt, VTK, OpenCV): raw continuous recording with acquisition timestamps, ROI depth time series, automatic σ, delivered fps and fill rate, and PLY/CSV export for cross-checking elsewhere. Then design and implement the breathing-tracking algorithm from ROI depth to respiratory waveform and gating signal, define the

TEAM RESULT

Consortium partners own 4DCT reconstruction, the image-guidance framework, and clinical advice. Programme-level results are not attributed to me.

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LIMITATIONS

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COLLABORATION BOUNDARY

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Recap and joint-development invitation

Build the optical part of surface-guided radiotherapy - patient-surface setup registration and respiratory gating - on a domestic 3D sensor stack.

PRIMARY CAPABILITY

3D Geometry & Registration / Sensor Fusion & Localization / Medical Navigation & Visualization

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

The clinical partner is the radiation oncology department of Seoul St. Mary's Hospital; interfaces are agreed with 4DCT reconstruction, image guidance, and integrated-control partners.

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